

Face Recognition System – Current Configuration Documentation

Overview

This document provides a concise, point-by-point description of the **exact face recognition system configuration** currently deployed. It covers models, embeddings, preprocessing, similarity metrics, and final decision logic.

1. Models

1.1 Face Detection

- **Model:** RetinaFace
- **Package:** Buffalo_L
- **Purpose:** Detects facial regions and landmarks within images

1.2 Face Recognition

- **Model:** ArcFace R50
- **Architecture:** ResNet-50–based
- **Package:** Buffalo_L
- **Purpose:** Generates discriminative facial embeddings

1.3 Library

- **Framework:** InsightFace
 - **Version:** v0.7.3
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2. Embeddings

- **Vector Size:** 512 dimensions
- **Normalization:** L2 normalization (unit vectors)
- **Data Type:** Float32

These embeddings are optimized for similarity comparison using cosine and correlation-based metrics.

3. Face Cropping & Alignment

3.1 Primary Cropping Method

- **Method:** landmark_alignment_5point
- **Technique:** Affine transformation using 5 facial landmarks
 - Left eye
 - Right eye
 - Nose
 - Left mouth corner
 - Right mouth corner
- **Image Source Handling:**

The system dynamically tracks the exact image variant (original, enhanced, or denoised) where the face was detected to ensure precise alignment.
- **Output Resolution:**
 - 112 × 112 pixels
 - Standard input size for ArcFace models

3.2 Fallback Cropping Method

- **Method:** bbox_crop_fallback
 - **Description:**
 - Simple bounding-box crop
 - Adds 20% padding around the detected face
 - **Usage Condition:**

Applied when landmark detection fails
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4. Similarity Metrics

The system computes multiple similarity scores for each face comparison. All scores are validated, and the strongest signal is selected.

4.1 Cosine Similarity

- **Range:** 0 – 1
- **Description:**
Primary similarity metric for face embeddings
- **Enhancement:**
Includes a quality boost for **CNIC vs. Selfie** comparisons

4.2 Pearson Correlation

- **Range:** 0 – 1
- **Description:**
Measures linear correlation between embedding vectors

4.3 Ensemble Score

A weighted combination of multiple similarity measures:

- **Cosine Similarity:** 0.6
- **Euclidean Distance:** 0.2
- **Pearson Correlation:** 0.2